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Question 1

Not answered

Marked out of 1.00

Question ID #14011: Haloperidol acts as a _____ receptor antagonist.

Select one:

- ☐ A. GABA
- ☐ B. norepinephrine
- ☐ C. dopamine
- ☐ D. glutamate

The correct answer is C.

Haloperidol is a widely prescribed antipsychotic medication that acts as a dopamine D2 receptor antagonist.

Answer A: GABA receptor antagonists are drugs that inhibit the action of GABA. In general, these drugs produce stimulant and convulsant effects and are mainly used for counteracting overdoses of sedative drugs. Examples include metrazol, flumazenil, and bicuculline.

Answer B: Beta-blockers are one type of medication that blocks norepinephrine and epinephrine (adrenaline) from binding to beta receptors on nerves. Other types of norepinephrine antagonists include tricyclic antidepressants and antipsychotics.

Answer D: Glutamate antagonists work by inhibiting the activity of glutamate receptors in the brain. These receptors are broadly classified into two main subtypes: N-methyl-D-aspartate (NMDA) and 3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) receptors.

Question 2

Not answered

Marked out of 1.00

Question ID #14012: Which area of the brain keeps track of the position of the body relative to the world?

Select one:

- ☐ A. primary motor cortex
- ☐ B. prefrontal cortex
- ☐ C. posterior parietal cortex
- ☐ D. supplementary motor cortex

The correct answer is C.

The posterior parietal cortex is one of the first areas of the brain to become active when planning a movement. People with damage to this area have trouble finding objects in space, even after describing their appearance accurately.

Answer A: The primary motor cortex is important for making movements but not for planning them.

Answer B: The prefrontal cortex, which is active during a delay before a movement, stores sensory information relevant to moving.

Answer D: The supplementary motor cortex is important for planning and organizing a rapid sequence of movements.

Question 3

Not answered

Marked out of 1.00

Question ID #14013: Ataxia is a common symptom of advanced multiple sclerosis and is characterized by:

Select one:

- ☐ A. impairment of movement
- ☐ B. an uncomfortable sense of restlessness or agitation
- ☐ C. impaired coordination and balance
- ☐ D. slow, writhing, involuntary movements of the extremities

The correct answer is C.

Ataxia is a symptom of several disorders including multiple sclerosis, cerebral palsy, and stroke. Ataxia involves a loss of balance and coordination that is caused by damage to the cerebellum, dorsal spinal cord, or vestibular system.

Answer A: A complete loss or significant impairment of movement is referred to as akinesia.

Answer B: An uncomfortable sense of restlessness or agitation is referred to as akathisia.

Answer D: Athetosis involves slow, writhing, involuntary movements.

Question 4

Not answered

Marked out of 1.00

Question ID #14014: Which of the following areas of the brain produces melatonin and plays a role in circadian and seasonal rhythms?

Select one:

- ☐ A. pineal gland
- ☐ B. suprachiasmatic nucleus
- ☐ C. reticular formation
- ☐ D. raphe nuclei

The correct answer is A.

The pineal gland is the primary endocrine organ that secretes hormones, such as melatonin, and controls the circadian rhythms.

Answer B: The suprachiasmatic nucleus (SCN) is a nucleus situated atop the optic chiasm. It contains a biological clock that is responsible for organizing many of the body's circadian rhythms.

Answer C: The reticular formation extends from the spinal cord through the hindbrain and midbrain into the hypothalamus. It is involved in respiration, posture, locomotion and REM sleep.

Answer D: The raphe nuclei are a group of nuclei located in the reticular formation of the medulla, pons, and the midbrain, situated along the midline that contains serotonergic neurons.

Question 5

Not answered

Marked out of 1.00

Question ID #14015: What term is defined as "the study of the relationship between physical stimulus magnitudes and their corresponding psychological sensations"?

Select one:

- ☐ A. parapsychology
- ☐ B. psychophysics
- ☐ C. pharmacokinetics
- ☐ D. synesthesia

The correct answer is B.

The question describes psychophysics. Several psychophysical laws have been developed including Weber's Law, Fechner's Law, and Steven's Power Law.

Answers A, C, and D: See rationale for Answer A.

Question 6

Not answered

Marked out of 1.00

Question ID #14016: The drug Ritalin inhibits the reuptake of which neurotransmitter?

Select one:

- ☐ A. serotonin
- ☐ B. dopamine
- ☐ C. GABA
- ☐ D. acetylcholine

The correct answer is B.

Ritalin (methylphenidate) blocks dopamine reuptake. Ritalin is used to treat children with ADHD.

Answer A: SSRIs block the reuptake of serotonin. Examples of SSRI medication include fluoxetine, escitalopram, and sertraline.

Answer C: Some examples of drugs that inhibit the action of GABA include securinine and metrazol.

Answer D: In the peripheral nervous system, acetylcholine (ACh) is released into the neuromuscular junction where it causes muscles to contract. In the central nervous system, ACh is involved in REM sleep, the regulation of the sleep-wake cycle, and learning and memory.

Question 7

Not answered

Marked out of 1.00

Question ID #14017: _____ is specialized for the control of fine movements, such as moving one finger at a time.

Select one:

- ☐ A. precentral gyrus
- ☐ B. prefrontal cortex
- ☐ C. caudate nucleus
- ☐ D. hippocampus

The correct answer is A.

The precentral gyrus is called the somatomotor cortex because it controls volitional movements of the contralateral side of the body.

Answer B: The prefrontal cortex is the brain region that has been implicated in planning complex cognitive behavior, personality expression, decision making, and moderating social behavior.

Answer C: The caudate nucleus functions not only in planning the execution of movement but also in learning, memory, reward, motivation, emotion, and romantic interaction.

Answer D: The hippocampus is a complex brain structure embedded within the temporal lobe. It has a major role in learning and memory.

Question 8

Not answered

Marked out of 1.00

Question ID #14018: A drug that binds with a receptor but does not activate it is called a _____.

Select one:

- ☐ A. indirect agonist
- ☐ B. receptor blocker
- ☐ C. indirect antagonist
- ☐ D. inverse agonist

The correct answer is B.

Receptor blockers are also known as direct antagonists. It is a type of ligand or drug that avoids or dampens a biological reaction. When binding to the receptor, it does not activate it, rather, it tends to block the particular receptor.

Answer A: An indirect agonist is a drug that attaches to a binding site on a receptor and facilitates the action of the receptor.

Answer C: An indirect antagonist is a drug that attaches to a binding site on a receptor and interferes with the action of the receptor.

Answer D: An inverse agonist produces an effect opposite to the effect produced by a neurotransmitter or an agonist.

Question 9

Not answered

Marked out of 1.00

Question ID #14019: Which structures are included in Papez's defined circuit?

Select one:

- ☐ A. mammillary bodies, basal ganglia, and circle of willis
- ☐ B. hippocampus, mammillary bodies, and thalamus
- ☐ C. hippocampus, cingulate gyrus, and hypothalamus
- ☐ D. putamen, cingulate gyrus, and amygdala

The correct answer is B.

James Papez originally proposed that the circuit connecting the hypothalamus to the limbic lobe mediates the experience and expression of emotion. It includes the hippocampus, mammillary bodies, anterior nuclei of the thalamus, and the cingulate gyrus.

Paul MacLean reconceptualized Papez's proposal and redefined the circuit to include connections in the forebrain, which includes the hypothalamus, amygdala, and septum. The question itself is asking about Papez's defined circuit and therefore answer B is the best response.

Answers A, C, and D: See rationale for Answer B.

Question 10

Not answered

Marked out of 1.00

Question ID #14020: According to General Adaptation Syndrome, the first stage of stress is classified as _____.

Select one:

- ☐ A. adrenaline surge
- ☐ B. resistance
- ☐ C. exhaustion
- ☐ D. alarm reaction

The correct answer is D.

Selye (1956) investigated physiological reactions to stress and determined that people respond to all types of stressful situations in the same manner. The first stage is called "alarm reaction."

Answer A: Although the first stage of stress is characterized by an increase of adrenaline and energy level, the name of the stage is "alarm reaction" not "adrenaline surge."

Answer B: The second stage of reaction to stress is "resistance." In this stage breathing and heart rate returns to normal but cortisol remains heightened.

Answer C: The final stage of prolonged stress is "exhaustion." Fatigue, depression, and illnesses occur.

Question 11

Not answered

Marked out of 1.00

Question ID #14021: The _____ maintains the body's internal homeostasis.

Select one:

- ☐ A. cingulate gyrus
- ☐ B. hypothalamus
- ☐ C. cerebellum
- ☐ D. adrenal medulla

The correct answer is B.

Homeostasis refers to the ability to maintain the internal environment of the body within limits that allow survival. A primary function of the hypothalamus is to maintain the body's homeostasis in addition to regulating temperature and blood pressure.

Answer A: The cingulate gyrus is part of the limbic system and is involved in processing emotions and behavior regulation.

Answer C: The cerebellum regulates motor movements and coordinates voluntary movements such as posture, balance, and coordination.

Answer D: The adrenal medulla is the innermost part of the adrenal gland and consists of cells that secrete epinephrine and norepinephrine.

Question 12

Not answered

Marked out of 1.00

Question ID #14022: Which disease is characterized by a combination of finger agnosia, right-left confusion, agraphia, and acalculia?

Select one:

- ☐ A. Gerstmann's syndrome
- ☐ B. Parkinson's disease
- ☐ C. Huntington's disease
- ☐ D. Wernicke-Korsakoff syndrome

The correct answer is A.

Gerstmann's syndrome results from damage to the parietal lobe and is characterized by the symptoms listed in the question.

Answer B: Parkinson's disease is a progressive disorder characterized by tremors, muscular rigidity, akathisia, and speech difficulties.

Answer C: Huntington's disease includes symptoms of anxiety, depression, apathy, fidgeting, facial grimaces, and forgetfulness.

Answer D: Wernicke-Korsakoff syndrome includes mental confusion, abnormal eye movements, ataxia, and amnesia.

Question 13

Not answered

Marked out of 1.00

Question ID #14023: A _____ refers to abnormal electrical activity in the brain.

Select one:

- ☐ A. Seizure
- ☐ B. Stroke
- ☐ C. Concussion
- ☐ D. Coma

The correct answer is A.

A seizure is due to abnormal electrical activity in the brain that causes symptoms including an aura (e.g. feeling, odor, or noise) signaling the onset of the seizure, loss of consciousness, and abnormal movement.

Answer B: Stroke is a sudden interruption in the blood supply of the brain.

Answer C: A concussion is a traumatic brain injury that results in headaches and problems with concentration.

Answer D: A coma is a deep state of prolonged unconsciousness.

Question 14

Not answered

Marked out of 1.00

Question ID #14024: Temporal lobe damage causes which of the following?

Select one:

- ☐ A. contralateral neglect
- ☐ B. apraxia
- ☐ C. receptive aphasia
- ☐ D. ataxia

The correct answer is C.

Damage to the temporal lobe may result in receptive aphasia (difficulty understanding spoken language), difficulty learning and retaining new information, in addition to disturbances in selective attention.

Answer A: Contralateral neglect involves a deficit in attention paid to one side of the visual field, usually the side that is contralateral to the damage.

Answer B: Apraxia is the inability to perform particular purposeful actions.

Answer D: Ataxia refers to the loss of muscle control and balance.

Question 15

Not answered

Marked out of 1.00

Question ID #14025: The _____ controls the flow of information between the spinal cord and the brain.

Select one:

- ☐ A. thalamus
- ☐ B. medulla
- ☐ C. cerebellum
- ☐ D. central nervous system

The correct answer is B.

The medulla is a structure in the hindbrain that controls the flow of information between the spinal cord and the brain. It also regulates a number of vital functions including heartbeat, breathing, and blood pressure.

Answers A, C, and D: See rationale for Answer B.

Question 16

Not answered

Marked out of 1.00

Question ID #14026: Which of the following disease is characterized by uncontrollable jerking movements, writhing movements, and dementia?

Select one:

- ☐ A. Huntington's disease
- ☐ B. Parkinson's disease
- ☐ C. Alzheimer's disease
- ☐ D. Tardive Dyskinesia

The correct answer is A.

Huntington's disease is a fatal inherited disorder that causes degeneration of the caudate nucleus and putamen. The main symptoms include uncontrollable jerking movements, writhing movements, and dementia.

Answer B: Parkinson's disease is a neurological disease characterized by tremors, rigidity of the limbs, poor balance, and difficulty in initiating movements.

Answer C: Alzheimer's disease is a degenerative brain disorder that causes progressive memory loss, motor deficits, and eventual death.

Answer D: Tardive dyskinesia is a movement disorder that can occur after prolonged treatment with antipsychotic medication, characterized by involuntary movements of the face and neck.

Question 17

Not answered

Marked out of 1.00

Question ID #14027: Sexual maturity occurs when gonadotropin-releasing hormone is secreted by which area of the brain?

Select one:

- ☐ A. hippocampus
- ☐ B. hypothalamus
- ☐ C. adrenal glands
- ☐ D. thalamus

The correct answer is B.

Sexual maturity occurs when the hypothalamus begins secreting a gonadotropin-releasing hormone, which stimulates the secretion of follicle-stimulating hormone and luteinizing hormone by the anterior pituitary gland.

Answer A: The hippocampus is part of the limbic system and plays a role in regulating motivation, emotion, learning, and memory.

Answer C: The adrenal glands are composed of the cortex and the medulla. They produce hormones that help regulate your metabolism, immune system, blood pressure, and responses to stress.

Answer D: The thalamus has extensive nerve connections to both the cerebral cortex and the midbrain. The main function of the thalamus is to relay motor and sensory signals to the cerebral cortex.

Question 18

Not answered

Marked out of 1.00

Question ID #14028: Which of the following describes a drug that opposes or inhibits the effects of a particular neurotransmitter on the postsynaptic cell?

Select one:

- ☐ A. receptor blocker
- ☐ B. direct agonist
- ☐ C. agonist
- ☐ D. antagonist

The correct answer is D.

An antagonist is a substance that blocks the action or effect of another substance.

Answer A: A receptor blocker is a drug that binds with a receptor but does not activate it.

Answer B: A direct agonist is a drug that binds with and activates a receptor.

Answer C: An agonist is a drug that facilitates the effects of a particular neurotransmitter on the postsynaptic cell.

Question 19

Not answered

Marked out of 1.00

Question ID #14029: Which disease is Levodopa (L-Dopa) most commonly used to treat?

Select one:

- ☐ A. Alzheimer's disease
- ☐ B. Parkinson's disease
- ☐ C. Huntington's disease
- ☐ D. Tourette Syndrome

The correct answer is B.

L-Dopa is the most effective and commonly used to treat Parkinson's disease due to its ability to cross the blood-brain barrier.

Answer A: Cholinesterase inhibitors are used to treat cognitive symptoms in Alzheimer's patients.

Answer C: Medications used to control movements (e.g., tetrabenazine) are associated with Huntington's disease.

Answer D: Medications that block dopamine are used to treat the symptoms of Tourette Syndrome.

Question 20

Not answered

Marked out of 1.00

Question ID #14030: Which of the following is directed toward the central nervous system, conveying sensory information?

Select one:

- ☐ A. dorsal root
- ☐ B. efferent axon
- ☐ C. afferent axon
- ☐ D. ventral root

The correct answer is C.

Afferent axons are incoming axons that bring sensory information towards the central nervous system.

Answer A: The dorsal root is the spinal root that contains incoming sensory fibers.

Answer B: Efferent axons are axons that are directed away from the central nervous system, conveying motor commands to muscles and glands.

Answer D: The ventral root is the spinal root that contains outgoing motor fibers.

Question 21

Not answered

Marked out of 1.00

Question ID #14031: Hearing a certain musical tone when looking at a specific color is an example of _____.

Select one:

- ☐ A. synesthesia
- ☐ B. trichromatic theory
- ☐ C. opponent-process theory
- ☐ D. retinal disparity

The correct answer is A.

Synesthesia is a rare condition in which the stimulation of one sensory modality triggers a sensation in another sensory area. It is known as "joining senses." The cause is unknown but it is believed to be due to biological factors.

Answers B, C, and D: See rationale for Answer A.

Question 22

Not answered

Marked out of 1.00

Question ID #14032: Damage to the reticular activating system is most likely to cause which of the following?

Select one:

- ☐ A. deficits in explicit memory
- ☐ B. disruptions in the sleep-wake cycle
- ☐ C. impaired balance and coordination
- ☐ D. Kluver-Bucy syndrome

The correct answer is B.

The reticular activating system (RAS) is a complex, net-like system of nuclei and fibers that extends from the spinal cord into the brain. Damage to the RAS disrupts the sleep-wake cycle and can produce a permanent, coma-like state.

Answer A: Damage to the hippocampus is associated with deficits in explicit memory.

Answer C: Damage to the cerebellum would cause impaired balance and coordination.

Answer D: Kluver-Bucy syndrome is characterized by reduced fear, increased docility, compulsive oral behaviors, and altered dietary habits.

Question 23

Not answered

Marked out of 1.00

Question ID #14033: Which of the following stats is true regarding hypertension?

Select one:

- ☐ A. teens have higher rates than older adults
- ☐ B. general rates are higher for women over men
- ☐ C. Black communities have higher rates than White communities
- ☐ D. White communities have higher rates than Black communities

The correct answer is C.

The prevalence of hypertension is related to age, race, and gender. Black communities have higher rates than White communities.

Answer A: Older adults have higher rates of hypertension than younger adults.

Answer B: In general, rates are higher for men than women. However, among older adults and African Americans, they are higher for women than men.

Answer D: See rationale for Answer C.

Question 24

Not answered

Marked out of 1.00

Question ID #14034: "Start low and go slow" is a general rule for which of the following?

Select one:

- ☐ A. recovery from TBI
- ☐ B. prescribing drugs to older adults
- ☐ C. hormone replacement therapy
- ☐ D. prescribing drugs to children and adolescents

The correct answer is B.

Because of the increased risk for adverse drug effects, a general rule when prescribing drugs for older adults is to "start low and go slow" (i.e. begin with a low initial dose and increase gradually, if necessary).

Answers A, C, and D: See rationale for Answer B.

Question 25

Not answered

Marked out of 1.00

Question ID #14035: Which structure performs metabolic activities, providing the energy that the cell requires for all other functions?

Select one:

- ☐ A. nucleus
- ☐ B. ribosomes
- ☐ C. membrane
- ☐ D. mitochondria

The correct answer is D.

The mitochondria is the powerhouse of the cell and is responsible for cellular respiration and production of most ATP.

Answer A: The nucleus stores the cell's hereditary material (e.g., DNA) and it coordinates the cell's activities.

Answer B: Ribosomes are particles consisting of RNA and function to synthesize proteins.

Answer C: The membrane is a thin layer that forms the outer boundary of a living cell with its main function serving to protect the cell from its surroundings.

Question 26

Not answered

Marked out of 1.00

Question ID #14036: Compared to White individuals, _____ are more sensitive to side effects of neuroleptics, benzodiazepines, lithium, and some antidepressants.

Select one:

- ☐ A. Asians
- ☐ B. African-Americans
- ☐ C. Asians and African-Americans
- ☐ D. Native Americans

The correct answer is C.

Research confirms that compared to Whites, higher proportions of Asians, and to a somewhat lesser degree, African-Americans are slower or poorer metabolizers of specific isoenzymes. This means they are more sensitive to the side effects of certain drugs.

Answers A, B, and D: See rationale for Answer C.

Question 27

Not answered

Marked out of 1.00

Question ID #14037: Lesions in the right (nondominant) hemisphere of the cerebral cortex are most likely to cause which of the following:

Select one:

- ☐ A. exaggerated reactions to benign events
- ☐ B. indifference or excessive cheerfulness
- ☐ C. excessive fear and anger
- ☐ D. decreased talkativeness

The correct answer is B.

Concerning emotions, the left hemisphere of the brain mediates positive emotions, while the right hemisphere mediates negative emotions. Lesions to the right hemisphere produce an opposite effect (e.g., indifference, emotional lability, and atypical humor).

Answer A: Exaggerated reactions to small events is associated with damage to the left hemisphere of the brain.

Answer C: Excessive fear and anger would be appropriate emotions regardless of lesions.

Answer D: Lesions in the right hemisphere often increase talkativeness.

Question 28

Not answered

Marked out of 1.00

Question ID #14038: Which disease is due to atrophy and reduced thiamine in the thalamus?

Select one:

- ☐ A. Wernicke-Korsakoff syndrome
- ☐ B. Graves' disease
- ☐ C. hypoglycemia
- ☐ D. Kluver-Bucy syndrome

The correct answer is A.

Wernicke-Korsakoff syndrome is due to a thiamine deficiency and is usually the result of chronic alcoholism.

Answer B: Grave's disease (hyperthyroidism) is a result of hypersecretion of thyroxine in the thalamus.

Answer C: Hypoglycemia is caused by excess insulin production in the pancreas.

Answer D: Kluver-Bucy syndrome is a result of lesions in the amygdala and temporal lobes.

Question 29

Not answered

Marked out of 1.00

Question ID #14039: _____ is caused by an excessive accumulation of cerebrospinal fluid (CSF) in the ventricles of the brain.

Select one:

- ☐ A. Spina Bifida
- ☐ B. Meningitis
- ☐ C. Hydrocephalus
- ☐ D. Encephalitis

The correct answer is C.

Hydrocephalus is a chronic, neurological condition caused by an abnormal accumulation of cerebrospinal fluid (CSF), resulting in pressure on the brain.

Answer A: Spina Bifida is a birth defect that occurs when the spine and spinal cord do not form properly.

Answer B: Meningitis is an inflammation (swelling) of the protective membranes covering the brain and spinal cord.

Answer D: Encephalitis is inflammation of the brain that is most commonly caused by a viral infection.

Question 30

Not answered

Marked out of 1.00

Question ID #14040: The neurotransmitter acetylcholine is most responsible for:

Select one:

- ☐ A. muscular contraction
- ☐ B. mood regulation
- ☐ C. situational responsivity
- ☐ D. an inhibitory effect

The correct answer is A.

Acetylcholine is a neurotransmitter and plays a key role in signaling nerve and muscle coordination.

Answer B: Dopamine is a neurotransmitter that modulates thinking, feeling, and acting.

Answer C: Norepinephrine is a neurotransmitter that helps regulate the body's system in response to changing situational demands. It can impact mood, attention, dreaming, learning, and some autonomic functions.

Answer D: Serotonin exerts an inhibitory effect and impacts mood, hunger, arousal, sleep, aggression, and sexual activity.

Question 31

Not answered

Marked out of 1.00

Question ID #14041: Which limbic brain structure is important for spatial and explicit memory and the consolidation of declarative memories?

Select one:

- ☐ A. amygdala
- ☐ B. hypothalamus
- ☐ C. hippocampus
- ☐ D. cingulate cortex

The correct answer is C.

The hippocampus is a limbic system structure that is important for spatial and explicit memory and the consolidation of declarative memories.

Answer A: The amygdala is a limbic system structure but its primary role is related to emotions.

Answer B: The hypothalamus is not a part of the limbic system and its role is with hunger, thirst, sex, sleep, movement, and body temperature.

Answer D: The cingulate cortex is a limbic system structure but its primary role is with attention, emotion, and the experience of pain.

Question 32

Not answered

Marked out of 1.00

Question ID #14042: A process where messages within a neuron are transmitted from a neuron's dendrites to the end of its axon is called:

Select one:

- ☐ A. hyperpolarization
- ☐ B. polarization
- ☐ C. depolarization
- ☐ D. conduction

The correct answer is D.

Conduction is the term for the electrical process that moves messages through a neuron.

Answer A: Hyperpolarization is a change in a cell's membrane potential that makes it more negative.

Answer B: Polarization is the electrical gradient across a cell's membrane.

Answer C: Depolarization is the reduction in the level of polarization across a cell's membrane.

Question 33

Not answered

Marked out of 1.00

Question ID #14043: Which of these drugs is not a benzodiazepine?

Select one:

- ☐ A. Valium
- ☐ B. Lorazepam
- ☐ C. Nardil
- ☐ D. Ativan

The correct answer is C.

Nardil is an MAOI, not a benzodiazepine.

Answers A, B, and D: These are all included in the benzodiazepine drug category.

Question 34

Not answered

Marked out of 1.00

Question ID #14044: Soon after starting an antidepressant, a woman develops several negative side effects including dry mouth, urinary retention, constipation, and blurred vision. Which of the following would alleviate these symptoms?

Select one:

- ☐ A. caffeine
- ☐ B. histamine agonist
- ☐ C. dopamine agonist
- ☐ D. cholinergic agonist

The correct answer is D.

The woman's symptoms are anticholinergic side effects. Cholinergic agonists (e.g., Nicotine) are drugs that lead to stimulation of cholinergic receptors. A cholinergic agonist would help alleviate anticholinergic side effects.

Answer A: Side effects of caffeine include insomnia, headaches, dehydration, and dizziness.

Answer B: A histamine agonist (e.g., Betazole) is a drug that causes increased activity at one or more of the four histamine receptor subtypes. Side effects include hives, itching, and dry mouth.

Answer C: A dopamine agonist (Mirapex) is a compound that activates dopamine receptors. This agonist can be helpful in relieving refractory nausea.

Question 35

Not answered

Marked out of 1.00

Question ID #14045: The hypothalamus is involved in all of the following functions except:

Select one:

- ☐ A. hunger and thirst
- ☐ B. sexual arousal
- ☐ C. sleep-wake cycle
- ☐ D. vision and audition

The correct answer is D.

The hypothalamus mediates a wide range of autonomic, endocrine, and behavioral functions including emotional states, the sleep-wake cycle, hunger/thirst, sexual behavior/reproduction, and temperature. However, it is not involved in vision or audition.

Answer A: The hypothalamus is the body's primary "thirst" center.

Answer B: Sexual arousal is influenced by the hypothalamus. It stimulates the pituitary gland to release sex hormones.

Answer C: The circadian rhythm is set by the suprachiasmatic nucleus of the hypothalamus, which regulates the sleep-wake cycle.

Question 36

Not answered

Marked out of 1.00

Question ID #14046: What is defined as "the practice of optimizing medication regimens by measuring specific drugs at designated intervals?"

Select one:

- ☐ A. practical clinical trials
- ☐ B. therapeutic drug monitoring
- ☐ C. oversight
- ☐ D. psychoactive monitoring

The correct answer is B.

Therapeutic drug monitoring (TDM) measures specific drugs at designated intervals to maintain a constant concentration in a patient's bloodstream. TDM is most frequently used for drugs with limited therapeutic ranges, drugs known to cause adverse effects, etc.

Answer A: Practical (pragmatic) clinical trials are randomized studies conducted in practice settings to evaluate the effects of drug interventions.

Answer C: See rationale for Answer B.

Answer D: Incorrect. See rationale for Answer B.

Question 37

Not answered

Marked out of 1.00

Question ID #14047: What is contraindicated for individuals with anxiety and tension, anorexia, psychosis, or Tourette's Disorder?

Select one:

- ☐ A. methylphenidate
- ☐ B. narcotics
- ☐ C. alcohol
- ☐ D. lithium

The correct answer is A. Methylphenidate is a psychostimulant that can cause decreased appetite, insomnia, dysphoria, abdominal pain, and tachycardia. It is specifically contraindicated for people with anxiety and tension, anorexia, severe hypertension, psychosis, recent drug or alcohol abuse, pre-existing tics, or family history of Tourette's Disorder.

Answers B, C, and D: Although these are likely not helpful for the individuals in question, methylphenidate is the only choice that is specifically contraindicated. See rationale for Answer A.

Question 38

Not answered

Marked out of 1.00

Question ID #14048: The offspring of someone with Huntington's disease has a ____ chance of developing the disorder.

Select one:

- ☐ A. 50%
- ☐ B. 20%
- ☐ C. 70%
- ☐ D. 30%

The correct answer is A.

Huntington's disease is a fatal inherited disorder that causes degeneration of the caudate nucleus and putamen. The main symptoms include uncontrollable jerking movements, writhing movements, and dementia. It has a 50% chance of being passed on to a biological child.

Answers B, C, and D: See rationale for Answer A.

Question 39

Not answered

Marked out of 1.00

Question ID #14049: Propranolol and other beta-adrenergic blocking drugs are most effective for:

Select one:

- ☐ A. reducing palpitations, tremors, and somatic complaints of anxiety.
- ☐ B. reducing psychic manifestations (worry, obsessive thoughts) of anxiety.
- ☐ C. reducing symptoms associated with major depressive disorder.
- ☐ D. increasing the somatic complaints and the psychic manifestations of anxiety.

The correct answer is A.

Beta-adrenergic blocking drugs (i.e., Propranolol) are used to treat several disorders including anxiety. Beta-blockers reduce somatic symptoms of anxiety but do not directly affect subjective manifestations.

Answer B: Though beta-blockers may indirectly reduce the subjective symptoms of anxiety, they are most effective in treating the physiological/somatic symptoms associated with anxiety.

Answer C: Beta-blockers slow the heart rate and lower blood pressure by blocking the effect of adrenaline. The most common adverse effect reported with use is depression.

Answer D: Beta-blockers have not been found to increase the somatic symptoms or the psychic manifestations related to anxiety.

Question 40

Not answered

Marked out of 1.00

Question ID #14050: As the result of an injury, Walter cannot recognize familiar objects by touch. Most likely, Walter has damage to which area of his brain?

Select one:

- ☐ A. parietal
- ☐ B. occipital
- ☐ C. temporal
- ☐ D. frontal

The correct answer is A.

Walter is exhibiting tactile agnosia as a result of damage to the parietal lobe. The parietal lobe contains the somatosensory cortex, which processes information from receptors in the skin that are responsible for detecting touch, proprioception (i.e., the position of the body in space), nociception (i.e., pain) and temperature.

Answer B: The occipital lobe is the visual processing center of the brain that contains the primary visual cortex. The occipital lobe is responsible for processing visual information.

Answer C: The temporal lobe is involved in auditory perception, processing sensory input, language comprehension, and emotion association. The primary auditory cortex is housed in the temporal lobe and receives sensory information from the ears.

Answer D: The frontal lobe is the part of the brain that is involved in motor function, problem-solving, memory, language, judgment, and impulse control.

Question 41

Not answered

Marked out of 1.00

Question ID #14051: Which antidepressant is likely to have an initially high sedative effect?

Select one:

- ☐ A. NDRI
- ☐ B. SSRI
- ☐ C. MAOI
- ☐ D. TCA

The correct answer is D.

TCAs are also known as tricyclics. These types of antidepressants block histamine receptors, which results in drowsiness and sedation.

Answer A: NDRIs (e.g. Wellbutrin) have fewer anticholinergic side effects than TCAs and do not cause sexual dysfunction but can aggravate pre-existing psychosis and seizures.

Answer B: SSRIs often have side effects of gastrointestinal disturbance, insomnia, tremor, frequent urination, and sexual dysfunction.

Answer C: The common side effects of MAOIs include anticholinergic effects, hypertensive crisis, weight gain and swelling, and headaches.

Question 42

Not answered

Marked out of 1.00

Question ID #14052: Which of the following is a long-term concern regarding the use of chlorpromazine?

Select one:

- ☐ A. orthostatic hypotension
- ☐ B. anterograde amnesia
- ☐ C. tardive dyskinesia
- ☐ D. ataxia

The correct answer is C.

Chlorpromazine is a traditional antipsychotic medication that is associated with several adverse side effects. The long-term use of chlorpromazine (i.e., Thorazine) can result in tardive dyskinesia, which are involuntary movements of extremities (e.g., lips, tongue, face, trunk).

Answer A: Orthostatic hypotension is a possible side effect of tricyclic antidepressants and is a drop in blood pressure when standing up from a sitting or lying position.

Answer B: Anterograde amnesia is a potential side effect of benzodiazepines.

Answer D: Ataxia is a potential side effect of certain medications, specifically barbiturates (i.e., Phenobarbital) and antiepileptics (i.e., Phenytoin).

Question 43

Not answered

Marked out of 1.00

Question ID #14053: Migraine headaches can be precipitated by or aggravated by which of the following?

Select one:

- ☐ A. smoking and blue light exposure
- ☐ B. obesity and increased sugar intake
- ☐ C. over-sleeping or extreme exercising
- ☐ D. menstruation and relaxation after stress

The correct answer is D.

Migraines are likely caused by a nerve that inflames the dura, spreading pain and changing local blood flow. Migraines are precipitated or aggravated by many factors including menstruation, stress, relaxation after stress, changes in barometric pressure, alcohol, decongestant overuse, and certain foods.

Answers A, B, and C: These factors have not been linked to migraines. See rationale for Answer D.

Question 44

Not answered

Marked out of 1.00

Question ID #14054: Which sensory information is not relayed by the thalamus into the cerebral cortex?

Select one:

- ☐ A. olfaction
- ☐ B. gustation
- ☐ C. vision
- ☐ D. audition

The correct answer is A.

The thalamus relays sensory impulses from receptors in various parts of the body to the cerebral cortex. The only sensory information that is not relayed by the thalamus is related to smell (olfaction). Incoming signals related to smell (olfaction) are sent to the olfactory bulb which then forwards the signals to other areas of the brain.

Answer B: Gustation (tasting) begins with the taste buds and ultimately is relayed to the gustatory cortex. The thalamus regulates the sensation of taste, receiving neuronal signals from various parts of the gustatory system.

Answer C: In the vision system, the thalamus receives input from the retina which is then sent to the primary visual cortex.

Answer D: The main auditory portion of the thalamus is called the medial geniculate body. Information is routed here and then relayed to the auditory cortex.

Question 45

Not answered

Marked out of 1.00

Question ID #14055: Long-term potentiation is believed to play a critical role in which of the following?

Select one:

- ☐ A. emotion and language
- ☐ B. language and thought regulation
- ☐ C. arousal and thought dysregulation
- ☐ D. learning and memory

The correct answer is D.

Long-term potentiation (LTP) is a physiological process that involves physical modification of nerve synapses, especially at glutamate receptors in the hippocampus. LTP is believed to play an important role in the synaptic changes that underlie learning and memory.

Answer A: Language appears to play a role in emotion. Though not associated with LTP, language helps individuals use concepts to create meanings of on-going sensory perceptions.

Answer B: Thought and language have not been associated with LTP. Thought has been found to come before language, while language has been termed an expression of thought.

Answer C: Arousal is not associated with LTP.

Question 46

Not answered

Marked out of 1.00

Question ID #14056: Which of the following is not true regarding gate-control theory?

Select one:

- ☐ A. cells in the spinal cord block incoming pain signals
- ☐ B. cells in the spinal cord open to incoming pleasure signals
- ☐ C. applying heat or cold can help "close the gate"
- ☐ D. the nervous system can only process a certain amount of sensory information at one time

The correct answer is B.

This is not an aspect of gate-control theory.

Answers A, C, and D: These are all aspects to the gate-control theory of pain. The theory states that "closing the gate" to pain signals reduces the pain someone experiences in a certain area.

Question 47

Not answered

Marked out of 1.00

Question ID #14057: The most common side effects consistent with lithium use are which of the following?

Select one:

- ☐ A. polyuria and polydipsia
- ☐ B. dysarthria and slurred speech
- ☐ C. renal failure and coma
- ☐ D. psychosis and muscle weakness

The correct answer is A.

Polyuria is a condition where the body urinates more than usual. Polydipsia is an abnormal thirst. Both of these symptoms are the most common side effect of taking lithium.

Answer B: Dysarthria and slurred speech can be severe side effects when lithium levels are in the above-average range.

Answer C: Renal failure and coma are rare side effects that can occur when lithium levels are above average.

Answer D: Psychosis and muscle weakness are severe side effects that can occur when plasma lithium levels are out of range.

Question 48

Not answered

Marked out of 1.00

Question ID #14058: Which of the following is relevant in hypoglycemia and diabetes?

Select one:

- ☐ A. thyroxine
- ☐ B. glucose
- ☐ C. ADH
- ☐ D. ACh

The correct answer is B.

Low blood glucose causes hypoglycemia and in patients receiving insulin for type 2 diabetes, a sudden increase in glucose levels can cause negative cognitive side effects. The reuptake and use of glucose in the body is mediated by insulin, which is released by the pancreas.

Answer A: Thyroxine is secreted by the thyroid gland and regulates metabolism.

Answer C: Antidiuretic hormone (ADH) is secreted by the pituitary gland. ADH acts in the kidneys to mediate fluid retention.

Answer D: Acetylcholine (ACh) is a neurotransmitter that is involved in muscle contractions, REM sleep, and memory.

Question 49

Not answered

Marked out of 1.00

Question ID #14059: If left untreated, hypertension can lead to _____.

Select one:

- ☐ A. heart failure, kidney failure, and stroke
- ☐ B. obesity and brain dysfunction
- ☐ C. paralysis and renal failure
- ☐ D. cognitive decline and eventual coma

The correct answer is A.

Primary hypertension accounts for 85-90% of all cases of high blood pressure. It is hard to diagnose because it is often without symptoms. But left untreated, it can lead to cardiovascular disease, heart failure, kidney failure, and stroke.

Answers B, C, and D: See rationale for Answer A.

Question 50

Not answered

Marked out of 1.00

Question ID #14060: Which of the following is not an anticholinergic effect?

Select one:

- ☐ A. blurred vision
- ☐ B. constipation
- ☐ C. increased urinary frequency
- ☐ D. confusion

The correct answer is C.

Increased urinary frequency is not an anticholinergic effect.

Answers A, B, and D. Anticholinergic effects are caused by several drugs including antipsychotics and tricyclic antidepressants. They include dry mouth, blurred vision, tachycardia, urinary retention, constipation, memory impairment, and confusion.

Question 51

Not answered

Marked out of 1.00

Question ID #14061: A _____ provides cross-sectional images of the brain by use of magnetic fields and radio waves.

Select one:

- ☐ A. fMRI
- ☐ B. CT scan
- ☐ C. PET
- ☐ D. MRI

The correct answer is D.

An MRI is a structural imaging technique, meaning it provides structural information about the brain. A MRI uses magnetic fields and radio waves.

Answer A: A fMRI is similar to an MRI but it is a functional imaging technique, which provides information on brain activity as opposed to structure.

Answer B: A CT scan is a structural imaging technique that uses X-rays.

Answer C: A PET is a functional neuroimaging technique that provides information on cerebral blood flow, glucose metabolism, and oxygen consumption. When using PET, a patient is injected with a radioactive tracer that is taken up by active brain cells.

Question 52

Not answered

Marked out of 1.00

Question ID #14062: What is defined as the examination of chromosomes and their abnormalities?

Select one:

- ☐ A. genetic testing
- ☐ B. biochemical testing
- ☐ C. cytogenetic testing
- ☐ D. molecular testing

The correct answer is C.

Cytogenetic testing is a type of genetic testing that examines chromosomes and their abnormalities.

Answer A: Although cytogenetic testing is a subtype of genetic testing, the question is clearly describing cytogenetic testing.

Answer B: Biochemical testing is a subtype of genetic testing. It involves examining the protein instead of the gene.

Answer D: Molecular testing is a subtype of genetic testing that examines DNA mutations.

Question 53

Not answered

Marked out of 1.00

Question ID #14063: What are common side effects of opioid use?

Select one:

- ☐ A. sweating, nausea, and constipation
- ☐ B. shortness of breath, diarrhea, and dizziness
- ☐ C. slurred speech, irritability, and impaired cognition
- ☐ D. dizziness, nausea, and rash

The correct answer is A.

Common symptoms of opioid (narcotic-analgesics) use include constricted pupils, increased perspiration, constipation, nausea, vomiting, and respiratory depression.

Answer B: These are common side effects of beta-blockers.

Answer C: These are common side effects of sedative-hypnotics.

Answer D: These are common side effects of using mood stabilizing drugs.

Question 54

Not answered

Marked out of 1.00

Question ID #14064: Which of the following are helpful in diagnosing seizures?

Select one:

- ☐ A. EEG, CT, and MRI
- ☐ B. EEG
- ☐ C. fMRI
- ☐ D. PET

The correct answer is A.

The diagnosis of a seizure disorder includes obtaining a medical history, a physical exam, an EEG, and sometimes an imaging technique such as CT or MRI.

Answers B, C, and D: This is a difficult question because although some of these techniques may be used to diagnose seizures (among other things), Answer A is the most thorough and correct response to the question.

Question 55

Not answered

Marked out of 1.00

Question ID #14065: Which sleep wave pattern is defined as "awake, rested, and relaxed"?

Select one:

- ☐ A. delta waves
- ☐ B. alpha waves
- ☐ C. beta waves
- ☐ D. theta waves

The correct answer is B.

Alpha waves occur when someone is awake, rested, and relaxed.

Answer A: Delta waves occur when someone is in deep sleep.

Answer C: Beta waves occur when someone is alert and in a fully awake state.

Answer D: Theta waves are defined by deep relaxation and light sleep.

Question 56

Not answered

Marked out of 1.00

Question ID #14066: How many women of reproductive age are regularly impacted by the symptoms of premenstrual syndrome (PMS)?

Select one:

- ☐ A. 1
- ☐ B. 10-30%
- ☐ C. 30-80%
- ☐ D. 40-60%

The correct answer is C.

Premenstrual syndrome (PMS) includes anxiety, depression, emotional lability, fluid retention, backache, food cravings, and headaches. Its cause is not well understood and reported estimates of prevalence range from 30-80% of women of reproductive age. 1.8-5.8% have symptoms that are severe enough to qualify for PMDD.

Answers A, B, and D: See rationale for Answer C.

Question 57

Not answered

Marked out of 1.00

Question ID #14067: The most dangerous side effect of MAOIs is hypertensive crisis. When can this occur?

Select one:

- ☐ A. MAOI combined with SSRIs
- ☐ B. MAOI combined with rigorous exercise
- ☐ C. MAOI combined with certain foods
- ☐ D. MAOI combined with dehydration

The correct answer is C.

Hypertensive crisis can occur when an MAOI is taken in conjunction with barbiturates, amphetamines, antihistamines, or certain other drugs, or with foods containing the amino acid tyramine (e.g. aged cheeses and meats, beer, red wine, chicken liver, avocados, bananas, etc.). Hypertensive crisis symptoms include headache, stiff neck, rapid heart rate, nausea, vomiting, sweating, and light sensitivity.

Answers A, B, and D: These have not been linked to hypertensive crisis. See rationale for Answer C.

Question 58

Not answered

Marked out of 1.00

Question ID #14068: In which of the following disorders does an elevated level of serotonin NOT play a role?

Select one:

- ☐ A. Schizophrenia
- ☐ B. Autism Spectrum Disorder
- ☐ C. Anorexia Nervosa
- ☐ D. Bulimia Nervosa

The correct answer is D.

Bulimia Nervosa (along with PTSD and OCD) is associated with low levels of serotonin. All of the other disorders are associated with elevated levels of serotonin.

Answers A, B, and C: These disorders are all associated with elevated levels of serotonin.

Question 59

Not answered

Marked out of 1.00

Question ID #14069: If a split-brain patient is shown a picture of apple pie to their right visual field, what is the patient capable of doing?

Select one:

- ☐ A. verbally identify "apple pie" but not be able to point to it
- ☐ B. point to apple pie but not verbally identify "apple pie"
- ☐ C. verbally name "apple pie" and point to it with their left hand
- ☐ D. verbally name "apple pie" and point to it with their right hand

The correct answer is D.

A study by Sperry and Gazzaniga (1967) found that a picture projected to the right visual field of a split-brain patient, the patient was able to name the object and pick it out with their right hand but not their left hand. This is because the information projected to the right visual field went to the left hemisphere only.

Answers A, B, and C: See rationale for Answer D.

Question 60

Not answered

Marked out of 1.00

Question ID #14070: Which of the following brain areas has not been implicated in the regulation of emotion?

Select one:

- ☐ A. amygdala
- ☐ B. hypothalamus
- ☐ C. corpora quadrigemina
- ☐ D. cerebral cortex

The correct answer is C.

The corpora quadrigemina are reflex centers involved with hearing and vision, therefore are not associated with the regulation of emotion.

Answer A: The amygdala plays a role in the perception and expression of anger, fear, sadness, happiness and attaches emotion to memories.

Answer B: The hypothalamus is involved in the translation of emotions into physical responses.

Answer D: The cerebral cortex governs positive emotions (left hemisphere) and negative emotions (right hemisphere).

Question 61

Not answered

Marked out of 1.00

Question ID #14071: How does naltrexone treat alcohol use disorder?

Select one:

- ☐ A. blocks cravings and blocks reinforcing effects
- ☐ B. blocks reinforcing effects only
- ☐ C. blocks cravings only
- ☐ D. increases unpleasant reactions

The correct answer is A.

Naltrexone is an opioid receptor antagonist. It blocks the craving for alcohol and blocks the reinforcing effects of alcohol.

Answer B: This answer is partially correct, but Answer A is the best response.

Answer C: This is partially correct but Answer A is a better and more thorough answer.

Answer D: This answer describes disulfiram (Antabuse), a different anti-alcohol drug. Antabuse inhibits alcohol metabolism, causing an accumulation of unpleasant reactions such as nausea, vomiting, sweating, headache, and hypotension, all of which deter an individual from drinking alcohol.

Question 62

Not answered

Marked out of 1.00

Question ID #14072: Agrammatism, anomia, and articulation difficulties are produced by lesions in and around which area of the brain?

Select one:

- ☐ A. Wernicke's area
- ☐ B. Broca's area
- ☐ C. prefrontal cortex
- ☐ D. angular gyrus

The correct answer is B.

Broca's area is located in the left hemisphere and is associated with speech production and articulation. Lesions in this area produce agrammatism, anomia, and articulation difficulties.

Answer A: Wernicke's area is a critical language area in the posterior superior temporal lobe. This area is primarily involved in comprehension and language processing.

Answer C: The prefrontal cortex plays an important role in working memory function. It likely is also associated with control of working memory.

Answer D: The angular gyrus is located near the parietal lobe and the occipital lobe. It allows for the association of words, different images, sensations, and ideas.

Question 63

Not answered

Marked out of 1.00

Question ID #14073: Which of the following is not one of the categories used in the Glasgow Coma Scale?

Select one:

- ☐ A. confusion
- ☐ B. eye opening
- ☐ C. verbal utterances
- ☐ D. motor responses

The correct answer is A.

Confusion is not a category of the Glasgow Coma Scale.

Answers B, C, and D: The Glasgow Coma Scale is used to assess the level of consciousness in individuals with a brain injury. A numbered score is given in three categories: eye opening, verbal utterances, and motor responses. The total score ranges from 3 (worst possible score) to 15 (highest possible score) and is the sum of the response in each category.

Question 64

Not answered

Marked out of 1.00

Question ID #14074: This theory states that the eye has unique color receptors for each primary color.

Select one:

- ☐ A. trichromatic theory
- ☐ B. opponent-process theory
- ☐ C. pyramidal theory
- ☐ D. spectrum theory

The correct answer is A.

The trichromatic theory of color states that there are three types of color receptors that are each receptive to a different primary color (red, blue, or green) and all the other colors are produced by variations in the activity of these receptors.

Answer B: The opponent-process theory states that there are three bipolar receptors (red-green, yellow-blue, and white-black).

Answer C: Pyramidal theory is not a valid theory of color vision.

Answer D: Spectrum theory is not a valid theory of color vision.

Question 65

Not answered

Marked out of 1.00

Question ID #14075: In people who are described as "Type A," which characteristics are strongly associated with health problems?

Select one:

- ☐ A. irritation and hostility
- ☐ B. impatience and competitiveness
- ☐ C. criticalness and impatience
- ☐ D. cynicism and hostility

The correct answer is D.

Individuals who fit a Type A behavior pattern are highly competitive, achievement oriented, and tend to be hostile, easily irritated, and impatient. Studies have confirmed that in these individuals, being cynical or hostile is associated with health problems, especially coronary heart disease in males.

Answers A, B, and C: See rationale for Answer D.

Question 66

Not answered

Marked out of 1.00

Question ID #14076: 80 to 85% of individuals diagnosed with multiple sclerosis (MS) are initially categorized with this subtype.

Select one:

- ☐ A. relapsing-remitting type
- ☐ B. secondary progressive type
- ☐ C. autoimmune type
- ☐ D. chronic subthreshold type

The correct answer is A.

Most individuals who have MS are diagnosed *initially* with "relapsing-remitting type," which has alternating periods of relapse and remission.

Answer B: Most individuals with MS are later diagnosed with "secondary progressive type," which involves a gradual worsening of symptoms without distinct periods of relapse and remission.

Answer C: Autoimmune type is not a valid type of MS; however, MS is believed to be an autoimmune response triggered by genetic, viral, or environmental factors.

Answer D: Chronic subthreshold type is not a valid type of MS.

Question 67

Not answered

Marked out of 1.00

Question ID #14077: The _____ states that once an action potential is triggered in an axon, it is propagated, without decrement, to the end of the fiber.

Select one:

- ☐ A. all-or-none law
- ☐ B. rate law
- ☐ C. Hebb rule
- ☐ D. Organizational-Activational hypothesis

The correct answer is A.

The all-or-none law states that the strength by which a nerve or muscle fiber responds to a stimulus is not dependent on the strength of the stimulus. If the stimulus is above the threshold, the nerve or muscle fiber will either give a complete response or no response at all.

Answer B: The rate law is the principle that variations in the intensity of a stimulus, or other information being transmitted, in an axon are represented by variations in the rate the axon fires.

Answer C: The hypothesis proposed by Donald Hebb (known as Hebb rule) states that the cellular basis of learning involves the strengthening of a synapse that is repeatedly active when the postsynaptic neuron fires.

Answer D: The Organizational-Activational hypothesis states that steroid hormones permanently organize the nervous system during early development, which is reflected in adult male or female typical behaviors. In adulthood, the same steroid hormones activate, modulate, and inhibit these behaviors.

Question 68

Not answered

Marked out of 1.00

Question ID #14078: The inability to form memories for events that happened after brain damage is known as which of the following?

Select one:

- ☐ A. episodic memory
- ☐ B. retrograde amnesia
- ☐ C. procedural memory
- ☐ D. anterograde amnesia

The correct answer is D.

Anterograde amnesia is a loss of the ability to create new memories after the event that caused the amnesia, leading to a partial or complete inability to recall the recent past, while long-term memory, before the event, remains intact.

Answer A: Episodic memories are memories of single personal events, situations, and experiences.

Answer B: Retrograde amnesia is the loss of memory for events that occurred before the brain damage. The individual may be able to encode and memorize new things that occur after the onset.

Answer C: Procedural memory is the development of motor skills and habits. It is part of long-term memory that is responsible for knowing how to do things and how to perform certain procedures.

Question 69

Not answered

Marked out of 1.00

Question ID #14079: Fatigue, mood swings, nausea, and loss of bone mass are symptoms characteristic of which of the following?

Select one:

- ☐ A. hormone replacement therapy
- ☐ B. puberty
- ☐ C. menopause
- ☐ D. prostate cancer

The correct answer is C.

The decrease in estrogen levels during menopause leads to "hot flashes," fatigue, mood swings, nausea, loss of bone mass, reduced elasticity in the skin, and vaginal dryness.

Answer A: Hormone replacement therapy (HRT) is aimed at reducing these symptoms which are a part of menopause.

Answer B: Signs of puberty include the development of secondary sex characteristics, growth of pubic hair, acne, and growth spurts.

Answer D: Symptoms of prostate cancer are not typically covered on the EPPP, but they include frequent urination, blood in the urine, erectile dysfunction, fatigue, and weight loss.

Question 70

Not answered

Marked out of 1.00

Question ID #14080: This neurotransmitter is implicated in eating, seizures, anxiety disorders, motor control, vision, and sleep.

Select one:

- ☐ A. GABA
- ☐ B. glutamate
- ☐ C. serotonin
- ☐ D. dopamine

The correct answer is A.

GABA is an inhibitory neurotransmitter. Low levels of GABA are linked to anxiety disorders and the level of GABA is implicated in the other categories listed in this question.

Answer B: Glutamate is an excitatory neurotransmitter and plays a role in learning and memory.

Answer C: Serotonin exerts an inhibitory effect and impacts mood, hunger, temperature, sexual activity, arousal, sleep, aggression, and migraines.

Answer D: Dopamine is involved in personality, mood, memory, and sleep.

Question 71

Not answered

Marked out of 1.00

Question ID #14081: Which of the following best describes the dopamine hypothesis?

Select one:

- ☐ A. Schizophrenia is due to reduced sensitivity of dopamine receptors
- ☐ B. Schizophrenia is due to oversensitivity of dopamine receptors or excessive dopamine levels.
- ☐ C. Schizophrenia is due to reduced levels of dopamine
- ☐ D. Schizophrenia is due to a complete absence of dopamine

The correct answer is B.

According to the dopamine hypothesis, Schizophrenia is due to overactivity at dopamine receptors either as the result of oversensitivity of the receptors or excessive dopamine levels.

Answers A, C, and D: See rationale for Answer B.

Question 72

Not answered

Marked out of 1.00

Question ID #14082: Doxepin, Elavil, and Tofranil all belong to which category of drugs?

Select one:

- ☐ A. conventional antipsychotics
- ☐ B. atypical antipsychotics
- ☐ C. MAOIs
- ☐ D. TCAs

The correct answer is D.

These drugs are all TCAs.

Answers A, B, and C: See rationale for Answer D.

Question 73

Not answered

Marked out of 1.00

Question ID #14083: Tourette Syndrome has been most associated with a dysfunction in which of the following structures of the brain?

Select one:

- ☐ A. dentate gyrus
- ☐ B. pons
- ☐ C. basal ganglia
- ☐ D. mammillary bodies

The correct answer is C.

Tourette Syndrome is a neuropsychiatric disorder characterized by vocal and motor tics. Tics represent a symptom traditionally associated with a dysfunction of the basal ganglia leading to an excess of the neurotransmitter, dopamine.

Answer A: The dentate gyrus is part of the hippocampus and is involved in the formation of new episodic memories.

Answer B: The pons contains nuclei that relay signals from the forebrain to the cerebellum, along with nuclei that contribute to sleep, respiration, swallowing, bladder control, taste, eye movement, and posture.

Answer D: The mammillary bodies are connected to the hypothalamus and are involved in episodic memory.

Question 74

Not answered

Marked out of 1.00

Question ID #14084: Which general hormone is most effective in helping to restore sexual arousal for women following removal of their sex organs and/or after menopause?

Select one:

- ☐ A. androgen
- ☐ B. estrogen
- ☐ C. progesterone
- ☐ D. testosterone

The correct answer is A.

When women experience a loss of sexual desire following menopause or removal of their sex organs, androgen is effective in restoring sexual arousal and orgasm.

Answer B, C, and D: These hormones are implicated in sexual arousal and sexual cycles but the correct answer is A.

Question 75

Not answered

Marked out of 1.00

Question ID #14085: Which type of neuron is located entirely within the central nervous system?

Select one:

- ☐ A. sensory neuron
- ☐ B. interneuron
- ☐ C. motor neuron
- ☐ D. multipolar neuron

The correct answer is B.

Interneurons are the central nodes of neural circuits, enabling communication between sensory and/or motor neurons within the central nervous system (CNS). They play vital roles in reflexes, neuronal oscillations, and neurogenesis.

Answer A: Sensory neurons are neurons that detect changes in the external or internal environment and send information about these changes to the central nervous system.

Answer C: Motor neurons are neurons located within the central nervous system that control the contraction of a muscle or the secretion of a gland.

Answer D: A multipolar neuron is a neuron with one axon and many dendrites attached to its soma.

Question 76

Not answered

Marked out of 1.00

Question ID #14086: Which of the following is not one of the categories of the spinal cord?

Select one:

- ☐ A. coccygeal
- ☐ B. cervical
- ☐ C. thoracic
- ☐ D. trunk

The correct answer is D.

Trunk is not one of the categories of the spinal cord. The spinal cord is divided into five divisions. From top to bottom they are: cervical, thoracic, lumbar, sacral, and coccygeal.

Answers A, B, and C: These are all included divisions of the spinal cord.

Question 77

Not answered

Marked out of 1.00

Question ID #14087: Abrupt cessation of _____ can produce nightmares and an REM rebound.

Select one:

- ☐ A. beta-blockers
- ☐ B. benzodiazepines
- ☐ C. barbiturates
- ☐ D. narcotics

The correct answer is C.

Barbiturate use causes a decrease in REM sleep, and abrupt cessation can produce an REM rebound and nightmares.

Answer A: Abruptly discontinuing beta-blockers can cause sweating, palpitations, headache, and cardiac arrhythmia.

Answer B: Abrupt cessation of benzodiazepines can lead to seizures, depersonalization, panic, and stroke.

Answer D: Withdrawal symptoms of narcotics include insomnia and flu-like symptoms.

Question 78

Not answered

Marked out of 1.00

Question ID #14088: Which of the following is not true regarding traditional vs. atypical antipsychotics?

Select one:

- ☐ A. alleviation of positive symptoms of Schizophrenia vs. alleviation of positive and negative symptoms of Schizophrenia.
- ☐ B. faster onset vs. slower onset
- ☐ C. anticholinergic side effects vs. reduced extrapyramidal side effects
- ☐ D. affect serotonin vs. affect glutamate

The correct answer is D.

Both typical and atypical antipsychotics affect dopamine receptors but atypical antipsychotics also affect receptors for other neurotransmitters including serotonin and glutamate.

Answer A: Typical antipsychotics treat positive symptoms but not negative. Atypical antipsychotics treat both negative and positive symptoms.

Answer B: Typical antipsychotics have a faster onset of therapeutic effects while atypical antipsychotics have a slower onset.

Answer C: Both types of antipsychotics have anticholinergic effects and extrapyramidal side effects but extrapyramidal side effects are less common with atypical antipsychotics.

Question 79

Not answered

Marked out of 1.00

Question ID #14089: _____ are used to classify certain genes that influence various behavioral traits.

Select one:

- ☐ A. molecular genetic methods
- ☐ B. quantitative genetic methods
- ☐ C. twin method
- ☐ D. adoption method

The correct answer is A.

Molecular genetic methods are used in behavioral genetics to classify genes that influence certain behavioral traits. This research is done on both humans and animals.

Answer B: Quantitative genetic methods are used to determine the effect of genetic and environmental factors on individual differences of many traits, including behavioral traits.

Answer C: The twin method is a specific type of behavioral genetics research that relies on data from identical or fraternal twins.

Answer D: The adoption method is a semi-experimental design that compares nature vs. nurture.

Question 80

Not answered

Marked out of 1.00

Question ID #14090: This type of migraine has gastrointestinal symptoms and no aura.

Select one:

- ☐ A. classic
- ☐ B. common
- ☐ C. cluster
- ☐ D. tension

The correct answer is B.

88% of migraine sufferers have "common migraine," which does not have an aura but has gastrointestinal or other symptoms.

Answer A: About 12% of migraine sufferers experience "classic migraine," which starts with an aura.

Answer C: A cluster headache is not a classification of migraine. A cluster headache involves excruciating pain, usually behind one eye or temple, and occurs one or more per day over a two to three month period.

Answer D: A tension headache differs from a migraine. A tension headache involves non-throbbing pain, usually located on both sides of the head, face, or back of the neck.

Question 81

Not answered

Marked out of 1.00

Question ID #14091: Rapid breathing, cerebral hypoxia, chest pain, dizziness, and impaired concentration are all symptoms of _____.

Select one:

- ☐ A. hypertension
- ☐ B. fainting
- ☐ C. hyperventilation
- ☐ D. seizure

The correct answer is C.

Hyperventilation is a rapid deep-breathing attack that produces a drop in carbon dioxide, leading to respiratory alkalosis and cerebral hypoxia. It can be triggered by anxiety and its symptoms include those listed in the question.

Answer A: Hypertension occurs when high blood pressure occurs. It is often asymptomatic.

Answer B: Fainting occurs when the brain temporarily does not receive enough blood and consciousness is lost. One may feel dizzy prior to fainting but the other symptoms listed in the question fit better with Answer C.

Answer D: The onset of a seizure can have many different symptoms including but not limited to loss of consciousness, muscle spasms, falling, rapid eye movements, and teeth clenching.

Question 82

Not answered

Marked out of 1.00

Question ID #14092: Which area of the brain consists of the medulla, the pons, and the cerebellum?

Select one:

- ☐ A. brainstem
- ☐ B. forebrain
- ☐ C. midbrain
- ☐ D. hindbrain

The correct answer is D.

The hindbrain is the posterior part of the brain. It consists of the medulla, the pons, and the cerebellum.

Answer A: The brainstem extends from the top of the spinal cord into the center of the forebrain. The medulla and pons, the midbrain, and certain central structures of the forebrain constitute the brainstem.

Answer B: The forebrain is the most prominent part of the mammalian brain. It consists of the thalamus, hypothalamus, cerebral cortex, hippocampus, and basal ganglia.

Answer C: The midbrain is located in the middle of the brain, although in adult mammals it is dwarfed and surrounded by the forebrain. Major structures considered as part of the midbrain include the tectum, tegmentum, superior colliculus, inferior colliculus, substantia nigra inferior colliculus, and the substantia nigra.

Question 83

Not answered

Marked out of 1.00

Question ID #14093: Which brain region is in control of coordination, balance, and posture?

Select one:

- ☐ A. medulla
- ☐ B. pons
- ☐ C. cerebellum
- ☐ D. basal ganglia

The correct answer is C.

The cerebellum is on the hindbrain and is responsible for the control of motor activities.

Answer A: The medulla coordinates swallowing, coughing, sneezing, breathing, heartbeat, and blood pressure.

Answer B: The pons connects the two halves of the cerebellum and integrates movements between the right and left sides of the body.

Answer D: The basal ganglia help coordinate voluntary movement, among other functions.

Question 84

Not answered

Marked out of 1.00

Question ID #14094: Which of the following is not a side effect of a TBI?

Select one:

- ☐ A. temporary tremor
- ☐ B. disorientation
- ☐ C. impaired memory
- ☐ D. alteration of consciousness

The correct answer is A.

A temporary tremor is not a common consequence of a traumatic brain injury (TBI).

Answers B, C, and D: These are all common side effects of a traumatic brain injury.

Question 85

Not answered

Marked out of 1.00

Question ID #14095: _____ is defined as a sudden or gradual onset of neurological symptoms resulting from disruption in the blood supply to the brain.

Select one:

- ☐ A. cerebrovascular accident
- ☐ B. paralysis
- ☐ C. seizure
- ☐ D. concussion

The correct answer is A.

A cerebrovascular accident (CVA) is also known as a stroke. This occurs when there is a disruption to the blood supply in the brain that causes neurological consequences. Three major causes of stroke include thrombosis, embolism, and hemorrhage.

Answer B: Paralysis is the loss of function in part of the body.

Answer C: A seizure is caused by abnormal electrical activity in the brain.

Answer D: A concussion is a traumatic brain injury that affects brain function.

Question 86

Not answered

Marked out of 1.00

Question ID #14096: During which type of seizure does someone exhibit a "blank stare" with frequent eye blinking?

Select one:

- ☐ A. generalized
- ☐ B. partial
- ☐ C. absence
- ☐ D. tonic-clonic

The correct answer is C.

During an absence (petit mal) seizure, a person often shows a "blank stare" while blinking. These seizures involve brief loss of consciousness without prominent motor symptoms.

Answer A: Generalized seizures are a larger category including tonic-clonic and absence seizures. Even so, the description in the question fits best with absence seizures as opposed to the larger category of generalized seizures.

Answer B: A partial seizure begins in one side of the brain and affects one side of the body, although it can spread and become a generalized seizure.

Answer D: A tonic-clonic (grand mal) seizure includes stiffening of the body, shaking of the limbs, and depression or confusion with amnesia of the event.

Question 87

Not answered

Marked out of 1.00

Question ID #14097: Which of the following is not included in the peripheral nervous system?

Select one:

- ☐ A. somatic nervous system
- ☐ B. central nervous system
- ☐ C. parasympathetic nervous system
- ☐ D. sympathetic nervous system

The correct answer is B.

The central nervous system (CNS) is not included in the peripheral nervous system (PNS). The CNS includes the brain and the spinal cord.

Answers A, C, and D: The peripheral nervous system is broken up into the somatic nervous system and the autonomic nervous system. The autonomic nervous system is further divided into the parasympathetic and sympathetic nervous systems.

Question 88

Not answered

Marked out of 1.00

Question ID #14098: What is sexual dimorphism?

Select one:

- ☐ A. sexual differences in the body
- ☐ B. sexual changes in puberty
- ☐ C. sexual differences between male and females brains and organs
- ☐ D. sexual changes in menopause

The correct answer is C.

Sexual dimorphism research has confirmed that there are sex-related differences in the characteristics of the brain and organs between men and women, particularly in the corpus callosum, hippocampus, and SCN.

Answers A, B, and D: See rationale for Answer C.

Question 89

Not answered

Marked out of 1.00

Question ID #14099: Red-green colorblindness affects _____.

Select one:

- ☐ A. 2%-5% of females
- ☐ B. 8%-10% of males
- ☐ C. 8%-10% of females
- ☐ D. 12%-15% of males

The correct answer is B.

About 8-10% of the male population is affected by an inability to distinguish between red and green. Blue-yellow color blindness and complete color blindness is much less common.

Answers A, C, and D: See rationale for Answer B.

Question 90

Not answered

Marked out of 1.00

Question ID #14100: Procedural memory is processed by which area of the brain?

Select one:

- ☐ A. hippocampus
- ☐ B. amygdala
- ☐ C. thalamus
- ☐ D. cerebellum

The correct answer is D.

The cerebellum plays a role in the learning of procedural memory and motor learning.

Answer A: The hippocampus is where new and complex memories are encoded.

Answer B: The amygdala helps determine what memories to store, and it plays a part in determining where the memories are stored based on emotional responses.

Answer C: The thalamus plays an important role in working memory and sustaining short-term memory.

Question 91

Not answered

Marked out of 1.00

Question ID #14156: Which structure secretes antidiuretic hormone (ADH) and somatropin (growth) hormone?

Select one:

- ☐ A. endocrine gland
- ☐ B. pancreas
- ☐ C. thyroid gland
- ☐ D. pituitary gland

The correct answer is D.

The pituitary gland secretes several hormones including ADH and growth hormone and has a strong influence on the other endocrine glands. For this reason, it is known as the "master gland."

Answer A: Although endocrine glands are involved in hormone production, the question is specifically describing the pituitary gland.

Answer B: The pancreas releases insulin.

Answer C: The primary purpose of the thyroid gland is to regulate general metabolism by secreting thyroxine.

Question 92

Not answered

Marked out of 1.00

Question ID #14157: Which of the following brain structures is responsible for interhemispheric communication?

Select one:

- ☐ A. precentral gyrus
- ☐ B. arcuate fasciculus
- ☐ C. cingulate gyrus
- ☐ D. corpus callosum

The correct answer is D.

The corpus callosum is a wide, flat bundle of neural fibers beneath the cortex that connects the left and right cerebral hemispheres and facilitates interhemispheric communication.

Answer A: The precentral gyrus is located in the frontal lobe and contains the motor cortex.

Answer B: The arcuate fasciculus connects Wernicke's area to Broca's area.

Answer C: The cingulate gyrus is part of the limbic system and is involved in the processing and formation of emotions.

Question 93

Not answered

Marked out of 1.00

Question ID #14158: Stephanie was attacked in a parking lot and therefore has become fearful of parking lots, parking structures, or anything that reminds her of the attack. This detailed and significant emotional arousal depends on which of the following brain area?

Select one:

- ☐ A. amygdala
- ☐ B. hypothalamus
- ☐ C. hippocampus
- ☐ D. thalamus

The correct answer is A.

The amygdala is responsible for fear of individual items and is involved in the recall of emotionally-charged experiences. This brain structure is important for learning what to fear.

Answer B: The hypothalamus plays a critical role in the release of hormones and regulating body temperature.

Answer C: The hippocampus is associated with learning and memory, rather than directly involved in emotions.

Answer D: The thalamus is involved in motor activity, language, and memory.

Question 94

Not answered

Marked out of 1.00

Question ID #14159: Which of the following neurotransmitters plays a role in forming long-term memories?

Select one:

- ☐ A. GABA
- ☐ B. glutamate
- ☐ C. norepinephrine
- ☐ D. serotonin

The correct answer is B.

Glutamate plays a role in learning, memory, and long-term potentiation (LTP) which forms long term memories.

Answer A: GABA affects eating, seizures, anxiety disorders, motor control, vision, and sleep.

Answer C: Norepinephrine plays a role in mood, attention, dreaming, learning, and some autonomic functions.

Answer D: Serotonin is implicated in mood, hunger, temperature regulation, sexual activity, aggression, sleep, and migraines.

Question 95

Not answered

Marked out of 1.00

Question ID #14160: Which of the following structures is associated with Huntington's disease, Parkinson's disease, Tourette's disorder, OCD, and ADHD?

Select one:

- ☐ A. thalamus
- ☐ B. cingulate cortex
- ☐ C. basal ganglia
- ☐ D. suprachiasmatic nucleus

The correct answer is C.

The basal ganglia are involved in planning, organizing, coordinating voluntary movement, and regulating motor actions. The disorders mentioned in this question are associated with dysfunction in the basal ganglia.

Answer A: The thalamus acts as a relay station, transmitting sensory information to the appropriate parts of the brain. Damage to this area is associated with Wernicke-Korsakoff syndrome.

Answer B: The cingulate cortex is involved in attention, emotion, and perception, and the experience of pain.

Answer D: The suprachiasmatic nucleus (SCN) mediates the sleep-wake cycle and other circadian rhythms. The SCN may be involved in seasonal affective disorder.

Question 96

Not answered

Marked out of 1.00

Question ID #14161: Damage at the _____ level results in paraplegia.

Select one:

- ☐ A. cervical
- ☐ B. thoracic
- ☐ C. lumbar
- ☐ D. sacral

The correct answer is B.

Paraplegia entails a loss of sensory and voluntary functioning in the legs. This occurs when there is damage to the thoracic region of the spinal cord.

Answer A: Damage to the cervical level of the spinal cord causes quadriplegia (tetraplegia), which involves a loss of functioning and sensory input in the arms and legs.

Answer C: Damage to the lumbar region of the spine does not cause paraplegia.

Answers C and D: Damage to the sacral region of the spine does not cause paraplegia.

Question 97

Not answered

Marked out of 1.00

Question ID #14162: What is associated with deprivation of REM sleep?

Select one:

- ☐ A. subsequent increased difficulty in entering REM
- ☐ B. no significant symptoms
- ☐ C. inability to recall information learned a long time ago
- ☐ D. inability to retain newly learned information

The correct answer is D.

People deprived of REM sleep often show impairment in the ability to retain new information.

Answer A: On the contrary, people who suffer from deprivation of REM sleep have what is known as a "REM rebound," spending more time in REM sleep.

Answer B: This is incorrect. See rationale for Answer D.

Answer C: Individuals who are deprived of REM sleep have difficulty remembering new information, not memories that were formed a long time ago.

Question 98

Not answered

Marked out of 1.00

Question ID #14163: The discrepancy between what the left and right eyes see, is termed which of the following?

Select one:

- ☐ A. retinal disparity
- ☐ B. strabismus
- ☐ C. astigmatism
- ☐ D. binocular

The correct answer is A.

Retinal disparity is a cue that humans use to perceive depth. It is the small difference between the images projected on the two retinas when looking at an object or scene.

Answer B: Strabismus (lazy eye) is a condition in which the eyes do not point in the same direction.

Answer C: Astigmatism is a blurring of vision for lines in one direction (e.g., horizontal, diagonal, vertical).

Answer D: Binocular means stimulation from both eyes.

Question 99

Not answered

Marked out of 1.00

Question ID #14164: The _____ are membranes that surround the brain and spinal cord.

Select one:

- ☐ A. ventricles
- ☐ B. meninges
- ☐ C. central canal
- ☐ D. hydrocephalus

The correct answer is B.

Meninges line the skull and vertebral canal and enclose the brain and spinal cord.

Answer A: Ventricles are structures that produce cerebrospinal fluid (CSF), and transport it around the cranial cavity.

Answer C: The central canal (ependymal canal) is a fluid-filled channel in the center of the spinal cord that transports nutrients.

Answer D: Hydrocephalus is a condition caused by an obstruction of the flow of CSF.

Question 100

Not answered

Marked out of 1.00

Question ID #14165: Which aphasia is characterized by the ability to talk but having nothing to say, inability to understand written and spoken language, but do have the ability to produce automatic responses (e.g. singing familiar songs, repeating phrases, etc.)?

Select one:

- ☐ A. transcortical motor aphasia
- ☐ B. mixed transcortical aphasia
- ☐ C. transcortical sensory aphasia
- ☐ D. conduction aphasia

The correct answer is B.

This question defines mixed transcortical aphasia. This occurs when Broca's and Wernicke's areas are both damaged.

Answer A: Transcortical motor aphasia occurs when the damage is only in Broca's area. It includes non-fluent, effortful speech, lack of spontaneous speech, and anomia.

Answer C: Transcortical sensory aphasia occurs with lesions only in Wernicke's area. It is characterized by deficits in comprehension, anomia, and fluent but meaningless speech with unimpaired repetition.

Answer D: Conduction (associative) aphasia occurs when the arcuate fasciculus (the area that connects Broca's area and Wernicke's area) is damaged. This disorder does not affect language comprehension but does result in impaired repetition, anomia, and paraphasia.

Question 101

Not answered

Marked out of 1.00

Question ID #14166: Which theory of emotion proposes that emotions represent perceptions of bodily reactions to sensory stimuli?

Select one:

- ☐ A. James-Lange theory
- ☐ B. Cannon-Bard theory
- ☐ C. two-factor theory
- ☐ D. cognitive-appraisal theory

The correct answer is A.

The James-Lange theory of emotions stresses the importance of peripheral factors and proposes that emotions are representations of bodily sensations.

Answer B: The Cannon-Bard theory places emphasis on the brain mechanisms that mediate emotion. It suggests that emotional and bodily reactions to stimuli occur simultaneously.

Answer C: The two-factor theory of emotion describes subjective emotional experience as a result of the combination of physiological arousal and cognitive interpretation of that arousal as well as the environment in which it occurs.

Answer D: Cognitive-appraisal theory proposes that emotions are universal but there are differences in the interpretation or appraisal of emotion-arousing events.

